

Acoustic JCAL Parameter Values for Polyurethane Foam

The Johnson-Champoux-Allard-Lafarge (JCAL) model describes the acoustical behavior of rigid-frame porous materials using up to six characteristic (non-acoustic) parameters: open porosity (ϕ), static airflow resistivity (σ or r), tortuosity (α_∞), viscous characteristic length (Λ), thermal characteristic length (Λ'), and static thermal permeability ($\bar{k}'_o$). The first five constitute the classical Johnson-Champoux-Allard (JCA) model; the sixth is added in the JCAL extension to more accurately capture low-frequency thermal dissipation [1]-[3]. Below is a synthesis of experimentally determined and numerically predicted JCAL parameter values for various polyurethane (PU) foams, drawn from the literature.

1. Fully Reticulated (Open-Cell) Polyurethane Foams

Fully reticulated PU foams have all cell windows open (reticulation rate $Rw = 100\%$) and typically exhibit high porosity, low tortuosity, and relatively low airflow resistivity.

Parameter	Symbol	Typical Range	Notes & References
Porosity	ϕ	0.968–0.99	Highly porous, typically ~ 0.98 [1], [4]
Airflow resistivity	σ ($\text{N}\cdot\text{s}\cdot\text{m}^{-4}$)	600–4,000	Increases as cell size decreases (500–1,500 μm); e.g., $\sim 1,677$ at $Cs \approx 1,000 \mu\text{m}$ [1]
Tortuosity	α_∞	~ 1.05	Approximately constant for fully reticulated foams [1], [4]
Viscous characteristic length	Λ (μm)	100–300	$\Lambda \approx t/4$ for fibrous-like struts (t = strut thickness); e.g., 128–240 μm for cell sizes 500–1,000 μm [1], [4]
Thermal characteristic length	Λ' (μm)	200–550	$\Lambda' \approx 0.56 \cdot Cs$; e.g., 250–500 μm [1], [4]
Ratio $n = \Lambda'/\Lambda$	n	~ 1.55	Mean value for fully reticulated foams [1]
Static thermal permeability	$\bar{k}'_o$ ($\times 10^{-9} \text{ m}^2$)	~ 4 –40	Typically larger than viscous permeability k_o ; e.g., sample A: $24.3 \times 10^{-9} \text{ m}^2$ [5]

Specific measured examples: For a fully reticulated foam with cell size $Cs = 673 \mu\text{m}$, Doutres et al. [4] report $\phi = 0.956$, $\sigma = 3,490 \text{ N}\cdot\text{s}\cdot\text{m}^{-4}$, $\alpha_\infty = 1.06$, $\Lambda = 187 \mu\text{m}$, $\Lambda' = 250 \mu\text{m}$. Ning et al. [5] provide additional examples: sample A ($\phi = 0.978$, $k_o = 6.78 \times 10^{-9} \text{ m}^2$, $\Lambda = 208.7 \mu\text{m}$, $\bar{k}'_o = 24.3 \times 10^{-9} \text{ m}^2$, $\Lambda' = 513 \mu\text{m}$, $\alpha_\infty = 1.05$); sample B ($\phi = 0.987$, $k_o = 9.77 \times 10^{-9} \text{ m}^2$, $\Lambda = 272 \mu\text{m}$, $\bar{k}'_o = 35.3 \times 10^{-9} \text{ m}^2$, $\Lambda' = 534.9 \mu\text{m}$); sample C ($\phi = 0.968$, $k_o = 4.57 \times 10^{-9} \text{ m}^2$, $\Lambda = 195.2 \mu\text{m}$, $\bar{k}'_o = 11.4 \times 10^{-9} \text{ m}^2$, $\Lambda' = 306.9 \mu\text{m}$).

2. Partially Reticulated Polyurethane Foams

Partially reticulated foams contain membranes that partially close cell windows (reticulation rate Rw w w); empirical relation: $\alpha_\infty = 1 + 0.5 \cdot (1 - Rw)$ [1] | Viscous characteristic length Λ (μm) | 20-200 | Decreases markedly with membranes; e.g., 24 μm for $Rw = 5\%$ [4] | Thermal

characteristic length | Λ' (μm) | 100-500 | Less affected than Λ ; depends on both cell size and closed-pore content [1] | Ratio $n = \Lambda'/\Lambda$ | n | 1.5-8.2 | Increases with closed-pore content; e.g., $n = 8.2$ for 80% closed pores [1] | Static thermal permeability | $\bar{k}'_o$ ($\times 10^{-9} \text{ m}^2$) | 1-40 | Generally decreases with closed-pore content [5] |

Specific measured examples [4]: Foam P2 ($Rw = 32\%$): $\phi = 0.958$, $\sigma = 17,440 \text{ N}\cdot\text{s}\cdot\text{m}^{-4}$, $\alpha_\infty = 1.73$, $\Lambda = 46 \mu\text{m}$, $\Lambda' = 220 \mu\text{m}$. Foam P3 ($Rw = 5\%$): $\phi = 0.971$, $\sigma = 19,360 \text{ N}\cdot\text{s}\cdot\text{m}^{-4}$, $\alpha_\infty = 2.16$, $\Lambda = 24 \mu\text{m}$, $\Lambda' = 458 \mu\text{m}$. Foam M10 ($Rw \approx 69\%$): $\phi = 0.982$, $\sigma = 3,670 \text{ N}\cdot\text{s}\cdot\text{m}^{-4}$, $\alpha_\infty = 1.25$, $\Lambda = 240 \mu\text{m}$, $\Lambda' = 310 \mu\text{m}$.

3. Key Microstructure-Parameter Relationships

The following semi-empirical scaling laws link measurable microstructural features to JCAL parameters for high-porosity PU foams [1], [6], [7]:

- **Porosity:** $\phi = 1 - C * q(t/l)^2$, where $C * q \approx 0.785$ for a triangular-concave strut cross-section, t = strut thickness, l = strut length. For a cell-size-based (2-parameter) model: $\phi = 1 - C * q/B^2$, where $B = l/t \approx 3.78$ [6].
- **Airflow resistivity** (fully reticulated): $\sigma = (C * \sigma \cdot \eta)/(\phi \cdot C^2 s)$, where $\eta = 1.85 \times 10^{-5} \text{ Pa}\cdot\text{s}$, $C * \sigma \approx 128 \cdot \alpha_\infty$, giving $\sigma \propto \text{Cs}^{-2}$ [6], [7].
- **Airflow resistivity** (partially reticulated): $\sigma = \sigma_0 \cdot [Rw]^{-1.1166}$, where σ_0 is the fully-reticulated value [1].
- **Thermal characteristic length:** $\Lambda' = [2Vf]/[As + (1 - Rw) \cdot Aw]$, where Vf = fluid volume, As = strut surface area, Aw = window area [1], [6].
- **Viscous characteristic length:** For fully reticulated: $\Lambda = \Lambda'/1.55$. For partially reticulated: $\Lambda = \Lambda'/n$ where $n = 1.55 \cdot [Rw]^{-0.676}$ [1].

4. The Role of the Sixth Parameter (Static Thermal Permeability)

The JCAL model adds the static thermal permeability $\bar{k}'_o$ to improve predictions of the dynamic bulk modulus at low frequencies. For PU foams, $\bar{k}'_o$ is typically 2-5 times larger than the viscous static permeability $k_o = \eta/\sigma$ [2], [5]. Measured values range from $\sim 1 \times 10^{-9} \text{ m}^2$ for highly resistive partially-reticulated foams to $\sim 40 \times 10^{-9} \text{ m}^2$ for fully reticulated, large-cell foams [5]. The relationship $\bar{k}'_o \approx \phi \cdot \Lambda'^2/8$ (derived for idealized pore shapes) is only approximate for real PU foams; therefore, direct experimental determination via the three-microphone impedance-tube method is recommended when using the full JCAL model [2], [3].

5. Comparison with Other Porous Materials

For context, typical JCAL parameter values for melamine foam (an open-cell foam with ligament-like structure) include: $\phi \approx 0.984$, $\sigma \approx 9,365 \text{ N}\cdot\text{s}\cdot\text{m}^{-4}$, $\alpha_\infty \approx 1.03$, $\Lambda \approx 116 \mu\text{m}$, $\Lambda' \approx 256 \mu\text{m}$, $\bar{k}'_o \approx 4.0 \times 10^{-9} \text{ m}^2$ in the uncompressed state [3]. PU foams span a much wider range of parameters, especially in airflow resistivity and tortuosity, due to the ability to control reticulation rate and cell size during manufacturing.

References

- [1] O. Doutres, N. Atalla, and K. Dong, "Effect of the microstructure closed pore content on the acoustic behavior of polyurethane foams," *Journal of Applied Physics*, vol. 110, no. 6, Sep. 2011. DOI:10.1063/1.3631021
- [2] O. Doutres, N. Atalla, and K. Dong, "A semi-phenomenological model to predict the acoustic behavior of fully and partially reticulated polyurethane foams," *Journal of Applied Physics*, vol. 113, no. 5, Feb. 2013. DOI:10.1063/1.4789595
- [3] L. Lei, N. Dauchez, and J. D. Chazot, "Prediction of the six parameters of an equivalent fluid model for thermocompressed glass wools and melamine foam," *Applied Acoustics*, vol. 139, pp. 44–56, Oct. 2018. DOI:10.1016/j.apacoust.2018.04.010
- [4] O. Doutres, M. Ouisse, N. Atalla, and M. Ichchou, "Impact of the irregular microgeometry of polyurethane foam on the macroscopic acoustic behavior predicted by a unit-cell model," *The Journal of the Acoustical Society of America*, vol. 136, no. 4, pp. 1666–1681, Oct. 2014. DOI:10.1121/1.4895695
- [5] J. Ning, G. Zhao, and X. He, "Non-acoustical parameters and sound absorption characteristics of porous polyurethane foams," *Physics of Fluids*, vol. 31, no. 3, Mar. 2019. DOI:10.1063/1.5079486
- [6] Y. Ji, S. Chen, and W. Zhu, "The effect of pore numbers in the cell walls of soybean oil polyurethane foam on sound absorption performance," *Applied Acoustics*, vol. 157, p. 107010, Jan. 2020. DOI:10.1016/j.apacoust.2019.107010
- [7] H. Abdessalam, B. Abbès, F. Abbès, Y. Li, and Y.-Q. Guo, "Prediction of acoustic properties of polyurethane foams from the macroscopic numerical simulation of foaming process," *Applied Acoustics*, vol. 120, pp. 129–136, May 2017. DOI:10.1016/j.apacoust.2017.01.021